

ML Engineer – CV test task [short]

Aerial vehicle detection · ~8 hours · thefourthlaw.ai

This is a small version of something our team handles every week - drone footage in, working detector out. We're curious how you'd approach the full pipeline from scratch: the way you scope, label, train and measure is what we'll read most closely. Aim for 8 hours and scope accordingly.

the videos

Five Pexels clips, free for commercial use, no attribution. Four for training, one held out for evaluation. The eval clip must not be touched during training, threshold tuning or model selection. If a link breaks, swap in a similar aerial vehicle clip and note it in the README.

Role	Scene	URL
train A	highway interchange	pexels.com/video/8968356
train B	rural highway, sparse traffic	pexels.com/video/5382494
train C	simple highway, top-down	pexels.com/video/8457857
train D	urban intersection	pexels.com/video/3405804
eval (held-out)	city highway, daytime	pexels.com/video/32179597

the approach

Roughly: auto-label train and eval videos and train an object detection model to detect vehicles. Single class - call it *vehicle* - is fine. If the eval clip doesn't cover both distance bands, add more clips to your eval set.

the four metrics - across distance bands

Compute each metric separately for two distance bands: **0-200 m** and **200-400 m**. Distance can be estimated from each GT box's pixel size given a known reference (e.g. car length) and the camera's FOV - state your assumptions. Report in this format (example numbers - replace with yours):

Metric	0-200 m	200-400 m
Detection rate $TP / (TP + FN)$	0.72	0.41
Precision $TP / (TP + FP)$	0.85	0.62
False alarms / min $FP \times 60 / N_frames$	1.3	4.7
Time to first detection <i>seconds</i>	0.3	1.8

mAP@0.5 across both bands is a bonus.

how to submit

A GitHub repo (public, or private with access shared) with the runnable code, a README walking us through your strategy and the key decisions you made, the metrics table, a few example predictions on the eval clip (ideally with ground-truth boxes overlaid), and your trained weights or a link to them.

notes

What we care about: The pipeline is the deliverable, not the score - low metric values are fine if the pipeline is reliable and the numbers are honest. The auto-labeling step is the heart of the exercise, so don't shortcut it - no training on existing labeled aerial datasets (VisDrone, AU-AIR, UAVDT), and pseudo-labels should be model-generated. Light manual cleanup is fine and welcomed - just describe how you did it and what changed. We'd also like to see a short video of the detector running on the eval clip.

good luck!

Use the 8 hours however suits you - one session, or split as 2 x 4h. We'll have 30-45 minutes to review your submission, so keep the scope tight enough to be walked through in that window. Don't ship more than we can sensibly review.